CISPA
HELMHOLTZ CENTER FOR INFORMATION SECURITY

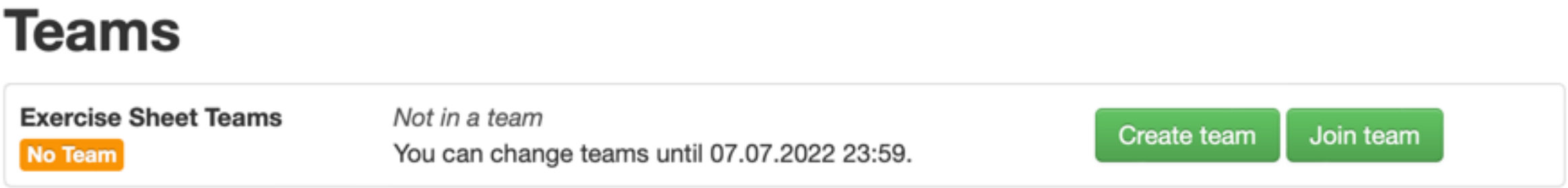
Cryptography

Lecture 03: Probability Theory

Nico Döttling

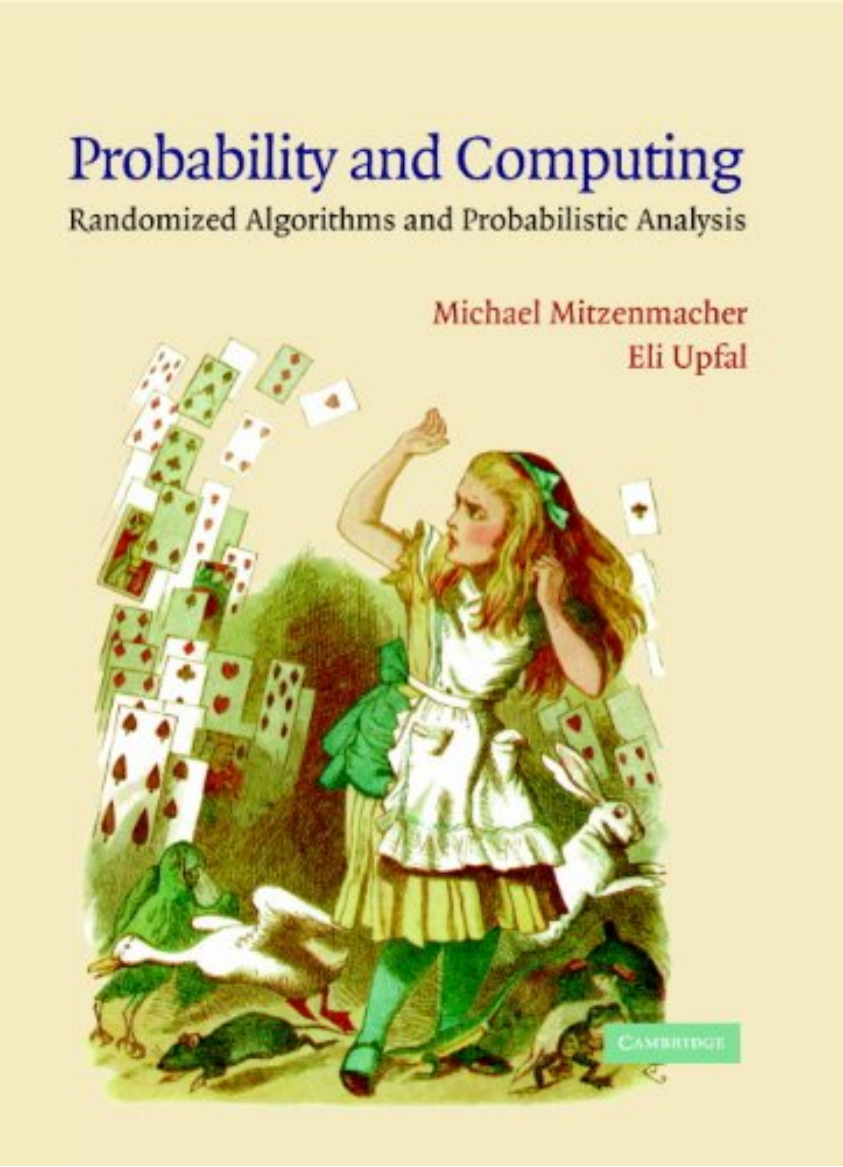
Announcement

- First exercise-sheet to be release on Friday
- Please register or join a team on CMS!



Literature

- Mitzenmacher and Upfal: Probability and Computing



Overview

- Probability Spaces, Events and Probabilities
- Conditional Probabilities
- Bayes' Rule
- Law of Total Probability
- Random Variables

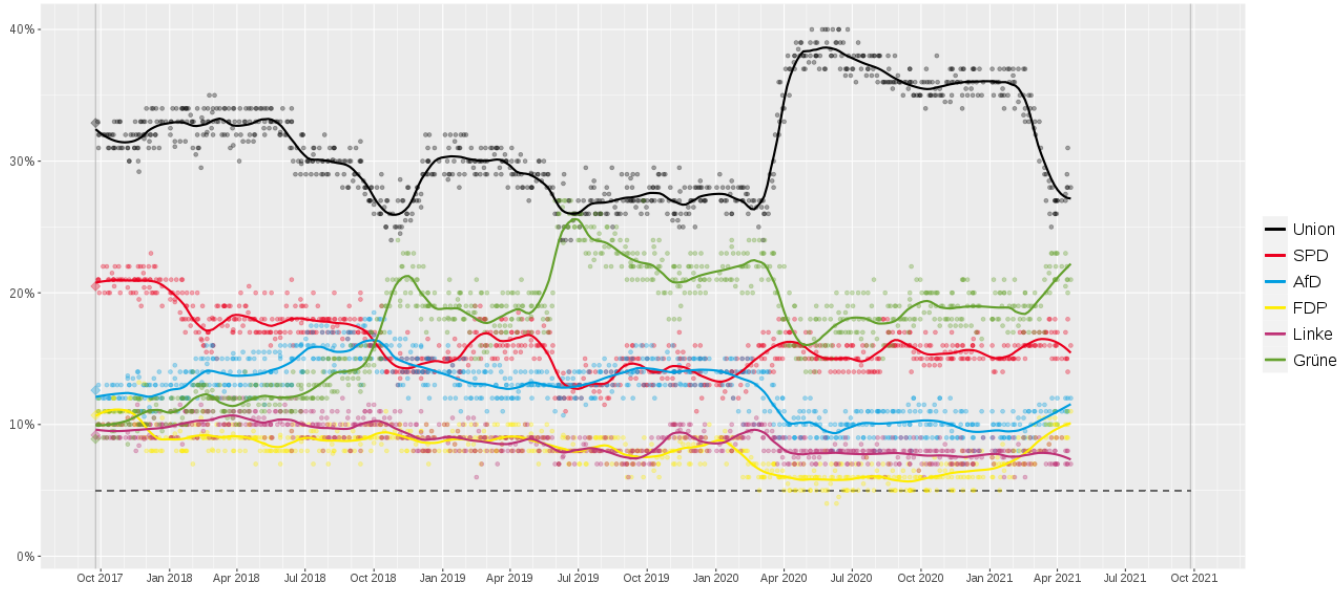
Why Randomness and Probability?

Probability Theory: Mathematical framework for systems which behave in unpredictable or random ways

- **Most Sciences:** Systems too complex for exact modeling, lack of precise measurements
- **Computer Science:** Randomness is a resource
- **Cryptography:** Impossible without randomness!

How can randomization lead to efficient algorithms?

- Example: Election Winners
- You have an election with 40 million voters
- You want to predict who the winner is without counting all votes
- Idea: Take a **random sample** of 10000 votes determine the winner!



Probability Spaces

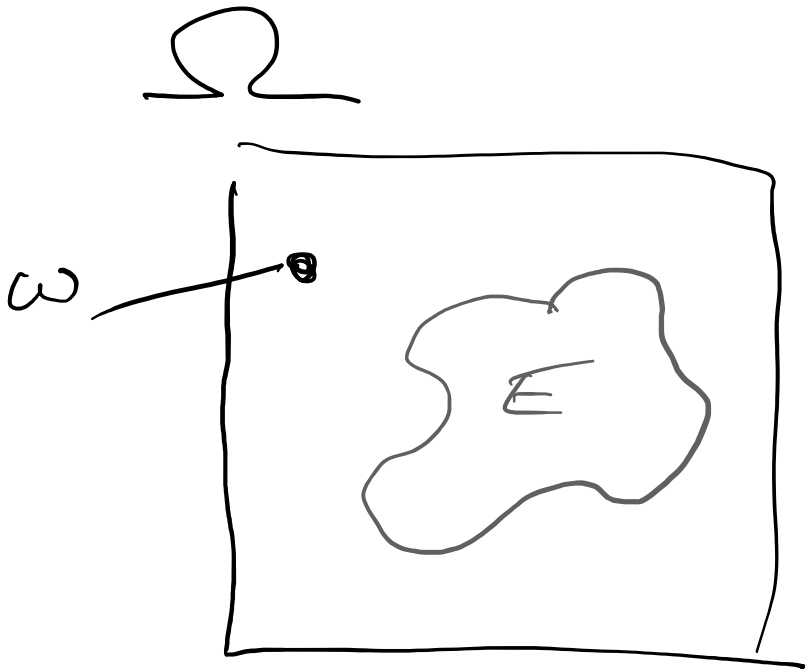
A finite probability space consists of two components:

1. A finite set Ω called the sample space.
2. A probability function $Pr: \Omega \to \mathbb{R}$ satisfying
   - For all $\omega \in \Omega \quad 0 \leq Pr[\omega] \leq 1$
   - $\sum_{\omega \in \Omega} Pr[\omega] = 1$

- The $\omega \in \Omega$ are called elementary events or outcomes
- A set $E \subseteq \Omega$ is called event
- For an event E define

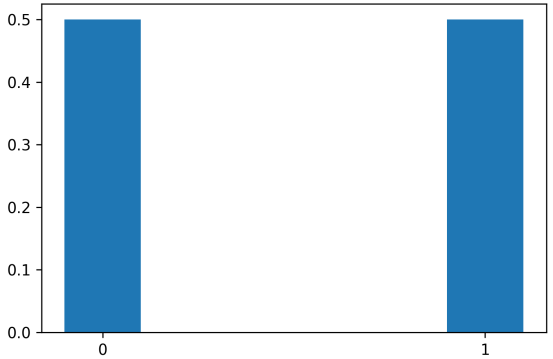
$$Pr[E] = \sum_{\omega \in E} Pr[\omega]$$

where $Pr[\emptyset] = 0$

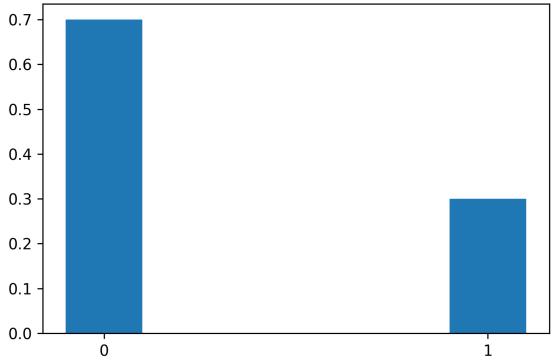


Examples

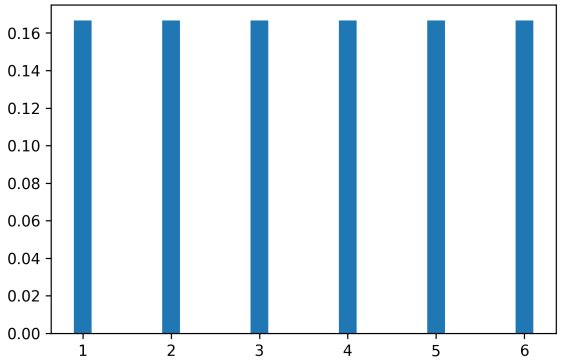
- Fair coin toss:
  $\Omega = \{0,1\}, Pr[0] = Pr[1] = 1/2$



- Biased Coin Toss:
  $\Omega = \{0,1\}, Pr[1] = p, Pr[0] = 1 - p$



- Fair Dice Roll:
  $\Omega = \{1,2,3,4,5,6\}, Pr[1] = Pr[2] = \cdots = Pr[6] = 1/6$



- Uniform randomness:
  $\Omega = \{1, \dots, N\}, Pr[1] = \cdots = Pr[N] = 1/N$

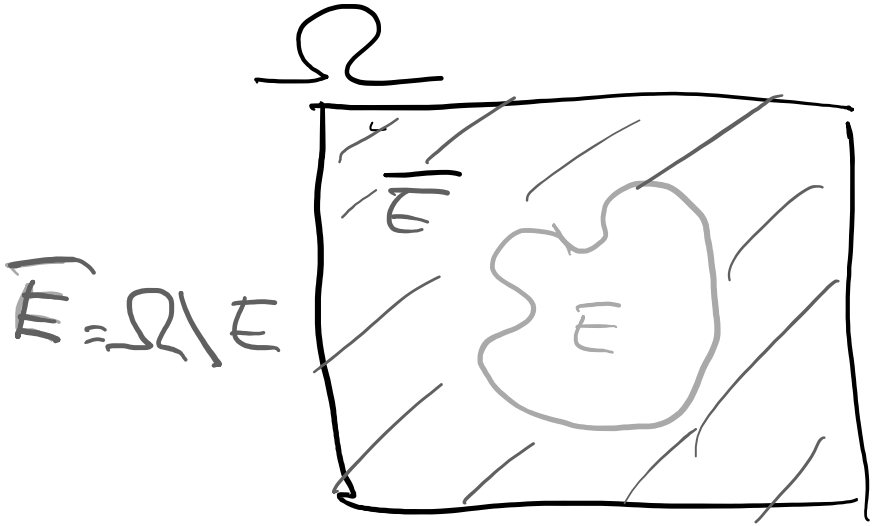
Set Operations on events

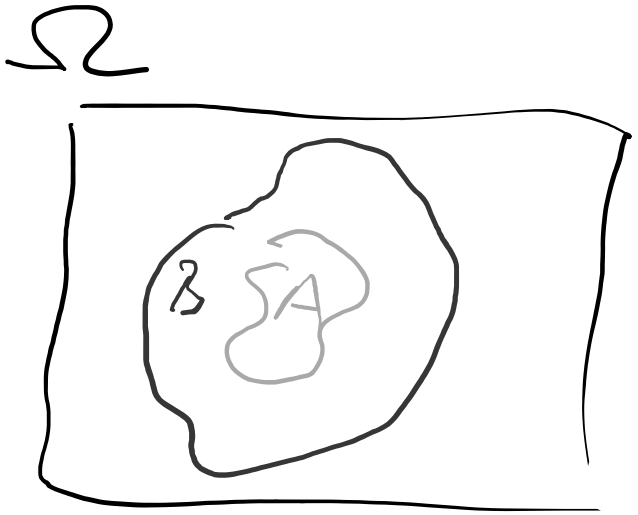
- The intersection $E_1 \cap E_2$ of two events E_1 and E_2 is interpreted as the logical "and" of two events
  - Depending on the context, we will use the following notations
    $Pr[E_1 \cap E_2] =: Pr[E_1, E_2] =: Pr[E_1 \wedge E_2]$
    $=: Pr[E_1 \, and \, E_2]$
- The union $E_1 \cup E_2$ is interpreted as the logical "or" of E_1 and E_2
  - Thus, we will use the notation
    $Pr[E_1 \cup E_2] =: Pr[E_1 \vee E_2] =: Pr[E_1 \, or \, E_2]$

Set Operations on events

- The complement $\overline{E} = \Omega \backslash E$ of an event E is interpreted as the logical "negation" of E
  - We will use the notation

    $$Pr[\overline{E}] =: Pr[\neg E] := Pr[notE]$$
- If $A \subseteq B$ we say the event A implies the event B and also write $A \Rightarrow B$



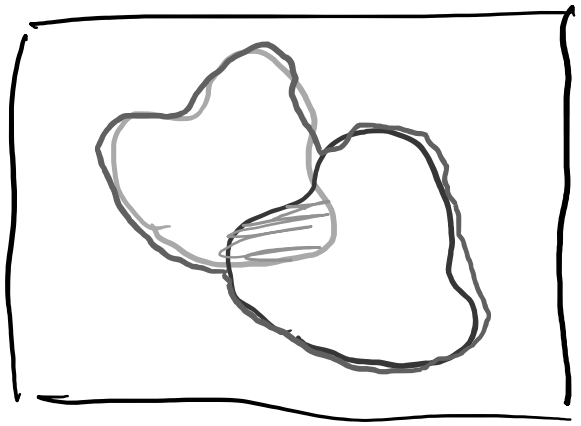


Properties of Probability Spaces

- For all events $A, B \subseteq \Omega$

1. $Pr[A \cup B] = Pr[A] + Pr[B] - \underbrace{Pr[A \cap B]}_{\geq 0}$
2. $Pr[\overline{A}] = 1 - Pr[A]$
3. If $A \subseteq B$ then $Pr[A] \leq Pr[B]$

≤ 0



- Item 1 implies the **union-bound**:
  For all $A, B \subseteq \Omega$ it holds $Pr[A \cup B] \leq Pr[A] + Pr[B]$

$$\Omega = A \cup \overline{A}$$

$$1 = Pr[\Omega] = Pr[A \cup \overline{A}]$$

$$= Pr[A] + Pr[\overline{A}] - \underbrace{\underbrace{Pr[A \cap \overline{A}]}_{\neq 0}}_{=0}$$

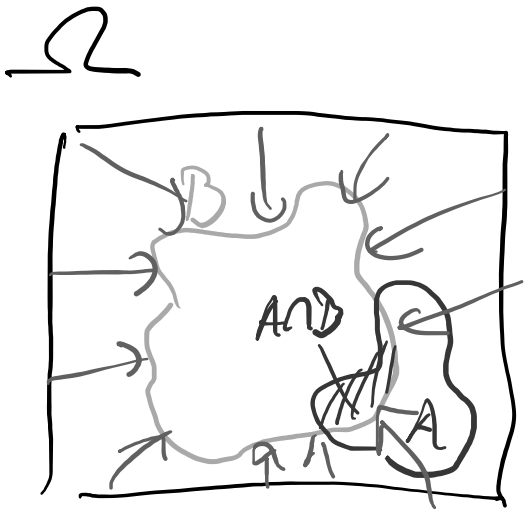
$$\Rightarrow Pr[\overline{A}] = 1 - Pr[A]$$

$$Pr[A_1 \cup A_2 \cup \ldots \cup A_n] \leq \sum_{i=1}^{n} Pr[A_i]$$

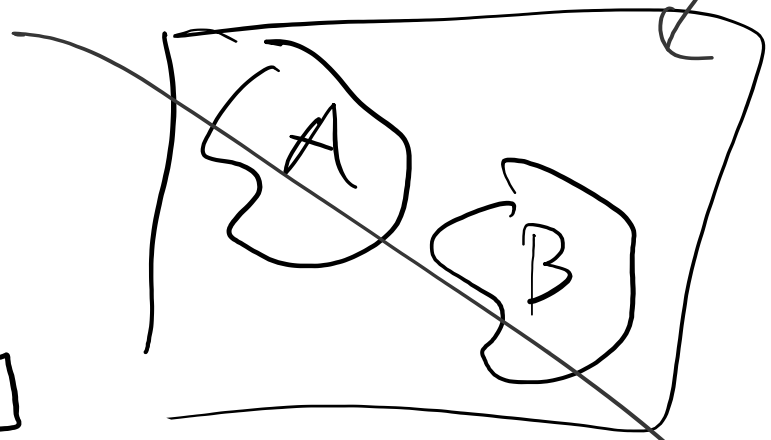
Conditional Probabilities and Independence

- For two events A and B $\neq \emptyset$ we define the conditional probability

$$Pr[A|B] = \frac{Pr[A \cap B]}{Pr[B]}$$



Disjointness



$A \cap B = \emptyset$

$Pr[B|A] = Pr[B]$

- Two events $A, B \subseteq \Omega$ are called independent if $Pr[A \cap B] = Pr[A] \cdot Pr[B]$

A, B independent

- This is equivalent to $Pr[A|B] = Pr[A]$
- By symmetry also to $Pr[B|A] = Pr[B]$

$$Pr[A|B] = \frac{Pr[A \cap B]}{Pr[B]} = \frac{Pr[A] \cdot Pr[B]}{Pr[B]} = Pr[A]$$

Example: Dice Roll

$$\Omega_1 = \{1, \dots, 6\} \qquad \Pr[1] = \dots = \Pr[6] = \frac{1}{6}$$

2 Dice rolls:

$$\Omega_2 = \{(1,1), (1,2), (1,3), \dots, (6,6)\} \qquad \Pr[(i,j)] = \frac{1}{36}$$

$$\Pr[\text{first dice roll is } 1] = \frac{1}{6}$$

$$\Pr[\underbrace{\text{second dice roll is } 2}_{E_2} \mid \underbrace{\text{first dice roll is } 1}_{E_1}] = \frac{\Pr[E_1 \cap E_2]}{\Pr[E_1]}$$

$$= \frac{\Pr[(1,2)]}{\Pr[E_1]} = \frac{\frac{1}{36}}{1/6} = \frac{6}{36} = \frac{1}{6}$$

$$\Pr[\text{first dice roll and second one have same outcome}]$$
$$= \Pr[(1,1)] + \Pr[(2,2)] + \dots + \Pr[(6,6)] = \frac{6}{36} = \frac{1}{6}$$

Bayes' Rule

- Bayes' rule relates the conditional probabilities $Pr[A|B]$ and $Pr[B|A]$
- This is especially useful when analyzing larger processes composed of smaller ones

Bayes' Rule

For all $\emptyset \neq A, B \subseteq \Omega$ it holds that $Pr[A|B] = \frac{Pr[B|A] \cdot Pr[A]}{Pr[B]}$

Proof:

$$Pr[A|B] = \frac{Pr[A \cap B]}{Pr[B]} \cdot \frac{Pr[A]}{Pr[A]} = Pr[B|A] \cdot \frac{Pr[A]}{Pr[B]}$$

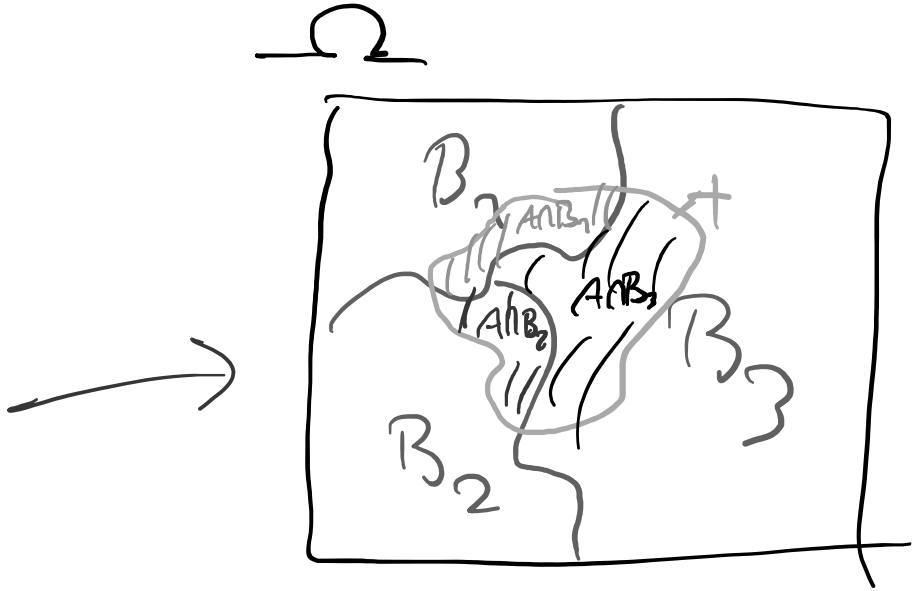
Law of Total Probability

- Sometimes we want to infer the probability of an event given conditional probabilities of this event
- This will also be useful to compute/bound probabilities by a **case analysis**

Law of Total Probability

Let $B_1, \dots, B_n$ be events that partition the probability space Ω, i.e. $B_1 \cup \dots \cup B_n = \Omega$ and $B_i \cap B_j = \emptyset$ for $i \neq j$. Then it holds for every event A that

$$\sum_{i=1}^{n} \underbrace{Pr[A|B_i] \cdot Pr[B_i]}_{=Pr[A \cap B_i]} = Pr[A]$$



Proof (Law of total probability)

If $E_i \cap E_j \neq \emptyset$ $(i \neq j)$ $\Rightarrow$ $\Pr[E_1 \cup \ldots \cup E_n] = \sum_{i=1}^{n} \Pr[E_i]$

$$\Rightarrow \underline{A = A \cap \Omega} = A \cap (B_1 \cup \ldots \cup B_n) = \underbrace{A \cap B_1}_{E_1} \cup \ldots \cup \underbrace{A \cap B_n}_{E_n}$$

$E_i \cap E_j \neq \emptyset$ (as $B_i \cap B_j \neq \emptyset$)

$$\Rightarrow \Pr[A] = \sum_{i=1}^{n} \Pr[E_i] = \sum_{i=1}^{n} \Pr[A \cap B_i] \cdot \frac{\Pr[B_i]}{\Pr[B_i]}$$

$$= \sum_{i=1}^{n} \Pr[A \mid B_i] \cdot \Pr[B_i]$$

$\square$

Example: Bayes and LotP in Action

- Assume 2 in 1000 people in a population have a certain disease
- There exists a test for this disease but it is not flawless
- If you have the disease, then there is a 95% chance that the test returns positive
- Moreover, if you don't have the disease there is a 3% chance that the test returns positive
- Imagine, you do a test and it returns positive, what is the probability that you actually have the disease?

Remarks

- The probability 2/1000 = 0.002 is your *prior* of having the disease, i.e. if you do not get the additional information of test, this is your best guess
- We are interested in the posterior probability~~, i.e. your~~

Example (cont.)

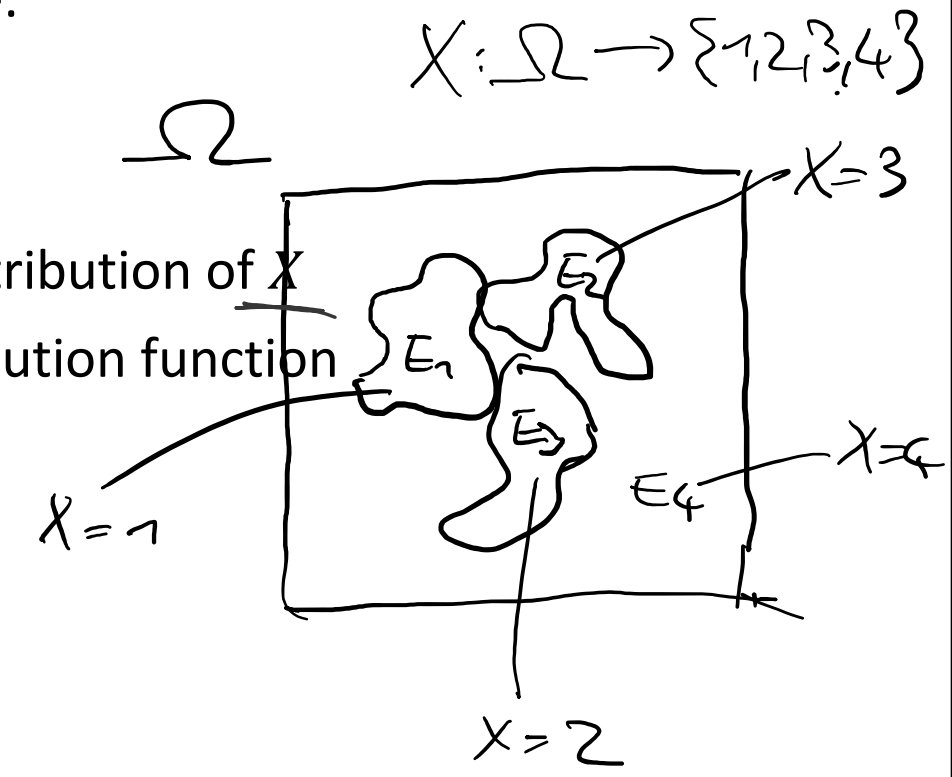
- Let's model this with a probability space
- Let A be the event that you have the disease. We know that $Pr[A] = 0.002$
- Let B be the event that the test returns positive. We know that $Pr[B|A] = 0.95$
- We also know that $Pr[B|\bar{A}] = 0.03$
- We want to know the posterior $Pr[A|B]$

$$Pr[A|B] \overset{\text{Bayes}}{=} \frac{\underbrace{Pr[B|A]}_{=0.95} \cdot \overbrace{Pr[A]}^{0.002}}{Pr[B]} = 0.06 = 6\%$$

$$Pr[B] \overset{\text{LofP}}{=} \underbrace{Pr[B|A]}_{0.95} \cdot \underbrace{Pr[A]}_{0.002} + \underbrace{Pr[B|\bar{A}]}_{0.03} \cdot \underbrace{Pr[\bar{A}]}_{1-Pr[A]=0.998} =$$

Random Variables

- In general, any function $X: \Omega \to D$ for some domain D is called a (D-valued) **random variable**
- We can now define events with respect to a random variable.
- Let $x \in D$ define the event $E_x = \{\omega \in \Omega: X(\omega) = x\} \subseteq \Omega$
- We write the shorthand "$X = x$" for E_x
- We call the function $F(x) = Pr[X = x]$ the probability distribution of X
- We often define random variables by their probability distribution function



Independence

- Consider two random variables $X \in D_X$ and $Y \in D_Y$
- We say that X and Y are **independent**, if it holds for all $x \in D_X$ and $y \in D_Y$ that the events $X = x$ and $Y = y$ are independent
- I.e. $Pr[X = x, Y = y] = Pr[X = x] \cdot Pr[Y = y]$ for all $x \in D_X$, $y \in D_Y$

Examples

- Identity function:
  $X: \Omega \to \Omega$. In this case $Pr[X = \omega] = Pr[\omega]$
- Projections:
  $\Omega \subseteq D_1 \times D_2 \quad X: \Omega \to D_1, (x, y) \mapsto x \quad Y: \Omega \to D_2, (x, y) \mapsto y$
- Reverse direction: Given two samples spaces Ω_1 and Ω_2, we can form a new sample space $\Omega = \Omega_1 \times \Omega_2$. Then the projections $X: \Omega \to \Omega_1$ and $Y: \Omega \to \Omega_2$ are independent random variables
- We say that a random variable $X: \Omega \to D$ is uniform on D or a uniform distribution, if for every $x \in D$ it holds $Pr[X = x] = 1/|D|$
- Sum of two dice rolls: $\Omega = \{1, \dots, 6\} \times \{1, \dots, 6\}$, $S: \Omega \to \mathbb{Z}$, $(x, y) \mapsto x + y$
- For an event E we can define the random variable $X_E: \Omega \to \{0,1\}$ such that $X_E(\omega) = 1$ if $\omega \in E$ and $X_E(\omega) = 0$ if $\omega \notin E$

Examples

Defining Probability Spaces via Random Variables

- From now on we will omit the function notation for random variables
- Moreover, from now on we will also omit defining the sample space Ω explicitly
- Instead, we implicitly define Ω via a set of random variables
- General idea: The probability space is defined via a few independent "base" random variables
- all other random variables of interest depend on these deterministically
- When constructing a system that uses randomization, we do not need to keep track of the sample space, it is define "on the fly" as we introduce new random variables
- If X is chosen uniformly random from some domain D, we will use the notation $X \leftarrow_{\$} D$ (instead of the function notation)

X is uniform on D_X

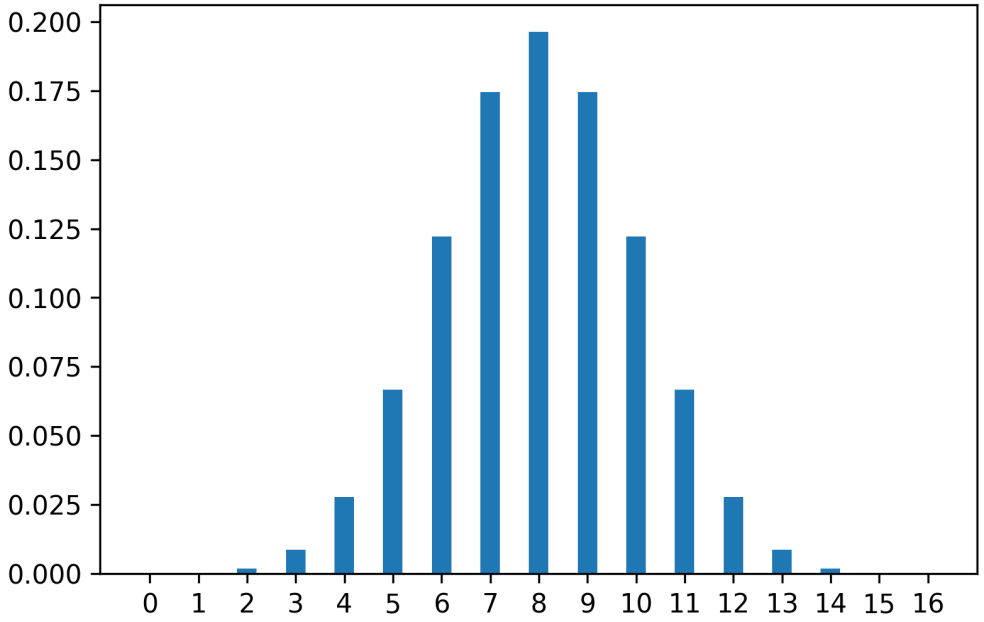
$\iff \Pr[X = x] = \frac{1}{|D_X|} \quad \forall x \in D_X$

Examples

- Let $X_1, \dots, X_n$ be independent random variables such that each X_i is uniform on $\{0,1\}$. We say $X_1, \dots, X_n$ are **iid uniform on** $\{0,1\}$ for short
- Then $X = (X_1, \dots, X_n)$ is uniform on $\{0,1\}^n$ Bernoulli
- $Y = X_1 + \cdots + X_n$ follow a distribution which is called the ~~Bernoulli~~ Binomial distribution

random coins

Central Limit Theorem

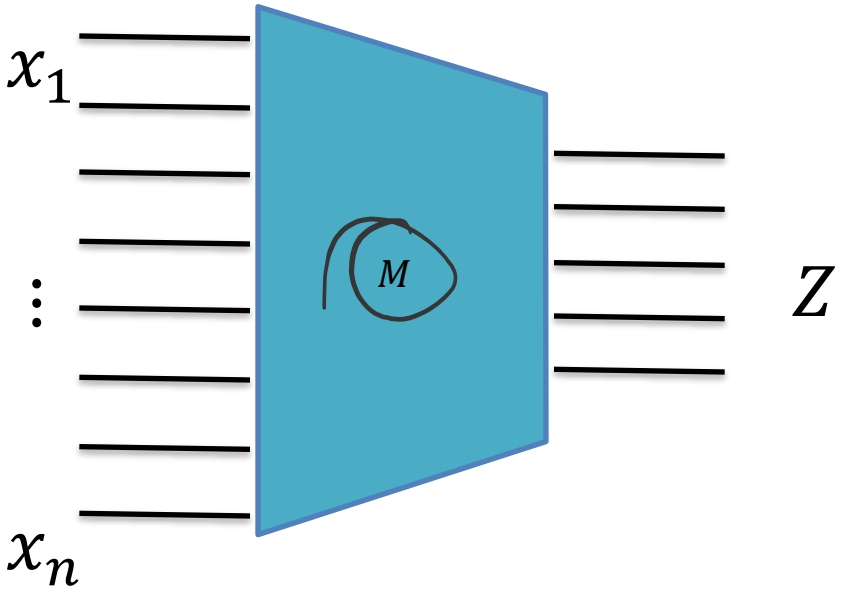


$$\Pr[Y = i] = \binom{n}{i} \cdot 2^{-n}$$

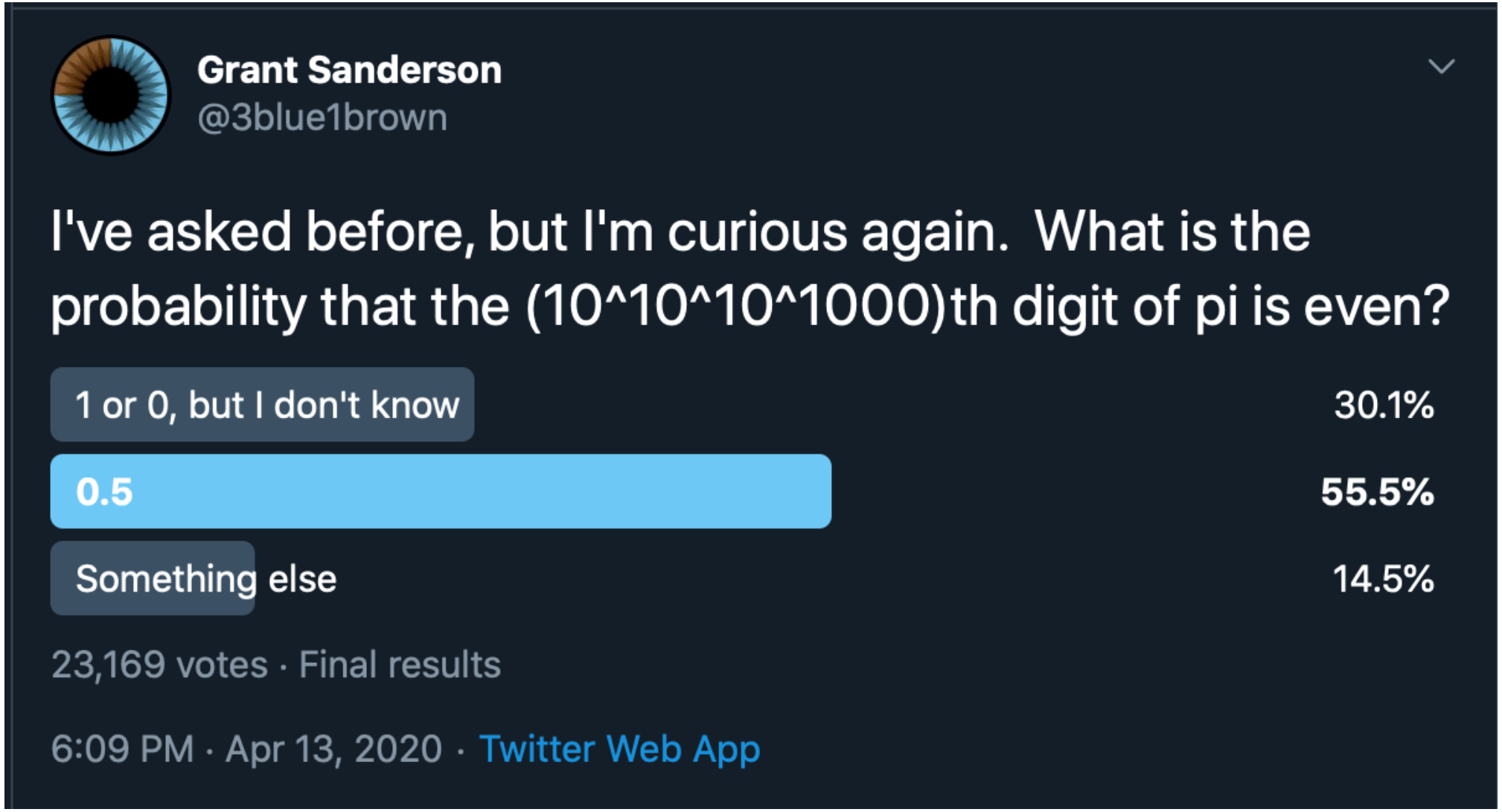
$i \in \{0, \dots, n\}$

Sampling Algorithms

- In general: We say a random variable Z is **sampleable**, if there exists an (efficient) algorithm M such that $Z = M(X_1, \dots, X_n)$, where $X_1, \dots, X_n$ are iid uniform on $\{0,1\}$
- $X_1, \dots, X_n$ are called the random coins or just coins of M



Summary



https://twitter.com/3blue1brown/status/1249731428322959361?s=21